

Tukul.com Construction Marketplace Specification

Tukul.com is a Malaysia-focused construction and renovation marketplace connecting property owners with verified contractors. It emphasizes **trust, transparency, and professionalism** throughout the project lifecycle. Owners post detailed project scopes, contractors bid with breakdowns, and payments are held in escrow and released only after each work phase is approved by the owner. This milestone-based workflow ensures accountability and resolves common industry issues (payment disputes, subpar work, unfinished projects) by embedding quality checks and secure payments into the platform flow ¹ ² .

Market Context and Competitive Landscape

Malaysian homeowners and contractors currently rely on informal networks or generic service apps for renovation projects. Platforms like **PilihKontraktor** position themselves as trusted marketplaces by verifying all contractors (SSM registration, certifications) and offering transparent, no-hidden-fees booking ³ . **RenSecure** targets home renovations by safeguarding payments: funds are held in trust and only released on approved milestones ⁴ . Tukul.com differentiates itself by combining competitive bidding with structured milestones. Unlike freelancer sites or Kaodim-style services, Tukul enforces written contracts with clear phase breakdowns, leveraging the Malaysian Standard Terms for Renovation Contracts (STCC-RSP 2015) to avoid disputes ⁵ . Its vision is akin to PilihKontraktor (“trusted platform for connecting customers with verified contractors” ⁶) but with stronger emphasis on project tendering, milestone management, and escrow.

Regulatory Environment (Malaysia)

- **Contractor Licensing:** Under Section 25 of the CIDB Act 1994, all contractors (even for small renovation jobs) must register with the Construction Industry Development Board (CIDB) before undertaking work ⁷ . Tukul should require proof of **CIDB registration** (G1–G7 license), SSM (business) registration, and any relevant safety or professional certifications for contractors. Integrating a CIDB check (via CIMS portal) ensures only licensed contractors bid. This aligns with PilihKontraktor’s practice of screening “every contractor with SSM, certifications, and proven experience” ⁸ .
- **Payment & Dispute Law:** Malaysia’s Construction Industry Payment and Adjudication Act (CIPAA 2012) mandates written contracts and adjudication for payment disputes ⁹ ¹⁰ . Notably, Section 35 of CIPAA voids “**pay-when-paid**” clauses: contractors cannot delay paying subcontractors simply because they haven’t been paid by the client ² . Tukul’s escrow model reinforces prompt, milestone-linked payments consistent with this law. Under CIPAA, all written construction contracts (including renovations and maintenance) are covered, so Tukul’s contracts will enable fast dispute resolution if needed ¹¹ ² .
- **Standard Contracts:** The CIDB has published the *Standard Terms of Construction Contract for Renovation and Small Projects (STCC-RSP 2015)*, a plain-language contract aiming to reduce disputes ⁵ . Tukul should incorporate key clauses from STCC-RSP into its agreement templates. For example, clauses on scope, variations, and payment schedule (milestones) can be adapted directly. Since STCC-RSP 2015 has no minimum or maximum project value limit ¹² , it fits Tukul’s range of small to large home projects.

User Roles and Permissions

1. **Property Owner (Client/Homeowner):** Can register and post projects, specify scope and requirements, review bids, select contractors, deposit payments, create and approve milestones, release funds, and rate completed work. Owners should view contractor profiles (with CIDB grade, portfolio, ratings) ⁸ ¹³ .
2. **Contractor:** A verified company or tradesperson who can browse projects, submit bids (lump-sum or phase-by-phase with breakdown), negotiate terms, manage milestones, and request payments on completion of each phase. Contractors have detailed profiles including **company details, CIDB grade, specializations, team size, past projects, licenses, and reviews** (much like PilihKontraktor's verified profiles) ⁸ . Only pre-approved contractors (via SSM/CIDB check) can bid.
3. **Platform Admin:** Oversees user registrations, verifies contractor credentials (CIDB, licenses, insurance), manages disputes or refund requests, and ensures compliance. Admins monitor project statuses, payment holds, and platform health.
4. **Optional Roles:** *Quality Inspector* (if integrated) who performs stage inspections, or *Project Manager* for projects that opt into full management (similar to RenSecure's optional add-ons ¹⁴).

Functional Requirements

- **Project Posting:** Owners create a project by filling a structured form (title, description, location, budget, photos). They attach detailed scope documents and can specify target start/end dates. They choose categories/trades (e.g. carpentry, plumbing). Owners may also select visibility (public or invite-only) for bids.
- **Bidding System:** Pre-registered contractors can place bids on open projects. Bids must include: total price (lump sum) *and optionally* a breakdown by work phases (milestones) with costs (labor/materials). This breakdown feeds the milestone builder. Contractors can upload their past work portfolio or similar project examples. The bidding interface guides contractors to outline all phases and share payment terms (deposit, stage payments). Contractors cannot see others' bids; owners see all bid proposals.
- **Milestone Builder:** Once a bid is accepted, Tukul provides an interactive milestone-scheduling tool. The selected contractor (with owner agreement) defines sequential phases (e.g. "Demolition", "Foundation", "Framing", "Finishing"), each with a description and agreed price. Each milestone forms a mini-contract stage. The owner approves the milestone plan. This **structured phase view** is visible to both parties. Final milestone typically includes a retention hold (e.g. 5–10% held for 1 month as warranty) to ensure quality ¹⁵ .
- **Escrow & Payments:** Tukul handles payments through a secure escrow system. The owner deposits the project's first payment into escrow (joint account or integrated payment gateway) before work starts. Subsequent milestone payments also go into escrow in advance of work. When a milestone is marked complete by the contractor, the owner reviews and either approves it or raises issues. Upon approval, Tukul releases that milestone's funds to the contractor. If the owner contests a stage, an internal dispute process (or optional CIPAA adjudication) can be initiated. A final holdback amount (released after a set period) covers any post-completion issues ¹⁵ . Tukul's terms ensure "*Payment is released to the contractor only when the property owner is fully satisfied*" ¹⁶ .
- **Secure Platform & Trust:** Tukul enforces "**Trust before transaction.**" Examples include: verifying contractor credentials (CIDB, SSM), requiring insurance details, having a review/rating system, and possibly partnering with a law firm to hold funds (as RenSecure does) ¹⁶ . All communications and transactions occur on-platform (no off-platform deals), with notifications at each step.

- **Reputation & Reviews:** After project completion, owners rate contractors on workmanship, communication, timeliness, etc. These ratings appear on contractor profiles. Likewise, contractors may rate owners for reliability and clarity. Over time, a reputation score builds for each user, making trust explicit (like PilihKontraktor’s “verified badge” concept) ¹⁷ ⁸ .
- **Search & Discovery:** Owners can browse contractor profiles by trade, location, CIDB grade, and ratings. Search filters include specialties, past project photos, or language spoken. Pinned high-rated contractors or those who frequently win bids can get featured listing.
- **Communication:** A built-in messaging/chat system allows owners and bidders to clarify project details before finalizing bids. Document attachments and questions can be exchanged securely.
- **Notifications:** Email/SMS alerts for key events: bid received, bid accepted, payment due, milestone ready for review, payment released, etc.
- **Mobile Compatibility:** A responsive website or mobile app so contractors can manage bids and milestones on the go.

Data Models

- **User/Account:** Common fields (name, email, contact, address). Role flag (owner or contractor). Login credentials. For contractors: business name, SSM ID, CIDB license (grade and expiry), tax ID, team size, insurance info, service categories, portfolio (projects, photos), ratings, response rate, and available locations. For owners: optionally saved property address(es), past projects, payment preferences.
- **Project Listing:** Unique ID, owner ID, title, detailed description, budget range, attachments, location, category tags, status (Draft, Open for Bids, In-Progress, Completed, Cancelled), timestamps.
- **Bid:** Bid ID, project ID, contractor ID, total price, optional detailed breakdown (as separate child items: milestone name, description, price, duration), date submitted, status (Pending, Selected, Rejected, Withdrawn), message/cover note.
- **Contract/Agreement:** Once a bid is accepted, a Contract record is generated: contract ID, project ID, contractor ID, owner ID, agreed total price, milestone schedule (see below), payment terms, date. The “contract” references the detail of milestones.
- **Milestone:** Milestone ID, contract ID, name, description, scheduled start/end dates, amount due, status (Not Started, In Progress, Completed, Approved, Payment Released), actual completion date.
- **Payment/Escrow:** Payment transaction records: payment ID, contract ID, milestone ID (if applicable), payer ID, payee ID, amount, timestamp, status (Deposited, Released, Refunded), type (Deposit, Stage Payment, Final Hold). These link to an external payment gateway or internal ledger.
- **Review:** Review ID, reviewer ID (owner), reviewee ID (contractor), project ID, rating (e.g. 1–5 stars) on categories, textual feedback, date.
- **Notification:** Records for queued notifications (email/SMS) linked to events.

APIs and Integration Points

While detailed API definitions depend on technical stack, key endpoints include:

- **Authentication API:** Register/login users (owners and contractors). OAuth or JWT tokens for session management.
- **User Profile API:** CRUD for user details, contractor verification status updates, rating retrieval.
- **Project API:** Create/read/update/delete (CRUD) for project listings. Endpoints to list open projects with filters (category, location, budget).
- **Bid API:** Submit a bid, view all bids for a project (owner view), accept/reject bid.

- **Milestone API:** Define/edit milestones once contract is formed, update milestone status (contractor marks complete, owner approves).
- **Escrow/Payment API:** Integrate with payment gateway for deposit and release. Endpoints to initiate deposit, mark milestone paid, refund. Possibly integrate a legal trustee account (like RenSecure uses a law firm's account) ¹⁸ or use a fintech escrow service.
- **Messaging API:** Send and retrieve messages between owner and bidders.
- **Notification API:** Manage sending alerts when events occur (bid accepted, payment processed, etc.).
- **Review API:** Submit and retrieve reviews, update contractor ratings.

Payment and Milestone Workflow

1. **Project Funding:** After a contractor bid is accepted, the owner makes an initial payment into Tukul's escrow (via online payment). This deposit covers the first milestone and secures the project. The system may allow splitting: e.g. 10–30% deposit and rest scheduled. (Consumer protection advice in Malaysia suggests low initial deposits ¹⁹.)
2. **Milestone Progress:** The contractor begins work on Milestone 1. Upon purported completion, the contractor marks the milestone “Complete” in Tukul. This triggers a notification to the owner to review.
3. **Owner Review:** The owner inspects the work (possibly aided by uploaded photos or an optional third-party inspection) and either **approves** or **requests revision**. If approved, Tukul releases the milestone payment from escrow to the contractor. If the owner disputes, Tukul's dispute resolution is triggered (see below).
4. **Subsequent Payments:** Steps 1–3 repeat for each milestone. The owner deposits each milestone's payment into escrow when the project was agreed; technically these funds could be held from the start or added before each stage. Either way, Tukul ensures funds are available before release.
5. **Final Holdback:** A predetermined retention (e.g. 5–10%) of the total contract value is held until project close-out. This final hold is released after a short defect liability period (e.g. one month) if no new issues arise ¹⁵.
6. **Completion:** Once all stages are approved and final hold released, the contract is complete. The owner submits a final review.

Dispute Resolution: If an owner refuses to approve a correctly completed milestone, Tukul's admin or appointed arbitrator reviews evidence (photos, communications). Failing that, parties may use formal adjudication under CIPAA (since Tukul's contract is in writing) ⁹ ¹⁰. Tukul may facilitate the CIPAA process (serve payment claim and response) but also encourages issue-raising early.

UI/UX Overview

- **Home/Landing Page:** Highlights Tukul's value (secure bidding, milestone payments, verified pros). A call-to-action for owners (“Post Your Project”) and contractors (“Create Contractor Profile”). Possibly a hero image of a renovation scene.
- **Post Project Form:** Step-by-step wizard: (1) Basic details (title, category, location); (2) Detailed scope (description, photos, docs); (3) Budget and timeline; (4) Confirmation. Use tooltips to guide owners on including necessary info.
- **Project Dashboard (Owner View):** Shows project status (Open for bids, Bidding closed, In Progress). Tabs for *Active Bids*, *Selected Contractor*, *Milestone Schedule*, *Messages*, *Payments*.
- **Contractor Profile Page:** Displays contractor's business info, CIDB grade, certifications, portfolio images, client reviews, and a “Verified” badge if applicable. Contact button or “Bid Now” if viewing from a project.

- **Bid Submission Screen (Contractor):** Form to enter total price, and a dynamic table to add milestone names, descriptions, and amounts. Show running total. Upload portfolio link or images. Option to write a short cover letter.
- **Milestone Builder (Contractor/Owner):** Visual timeline or list of milestones. Owners and contractors can add/edit phases before finalizing. Each phase has fields: name, description, due date, amount. The UI highlights total sum to match the bid amount.
- **Contractor Dashboard:** Lists awarded projects, upcoming milestones, and past projects. Each project entry shows current milestone, next payment due. Contractors can mark milestones as complete.
- **Payment Status:** A secure area showing escrowed funds, available balance (to contractor) or pending payments (for owner). Payment gateway integration for deposits and withdrawals.
- **Chat/Messaging Interface:** Threaded conversation for each project, with attachments. Could integrate real-time chat for convenience.
- **Notifications & Alerts:** In-app and email notifications list.
- **Mobile Responsiveness:** Key pages (dashboard, bidding, messaging, milestones) optimized for mobile.

(Note: If prototypes or wireframes exist, reference them here. Ensure form steps and navigation are consistent.)

Acceptance Criteria

To ensure quality and completeness, Tukul.com must meet these criteria:

- **Functional Completeness:** All features above are implemented and working as specified (project posting, bidding, milestone workflow, escrow payments, verification, reviews).
- **Usability:** The UI uses clear labels, guided forms, and logical flow. Users can find projects or contractors easily with filters and search. No form has excessive fields without tooltips.
- **Data Integrity:** Contractor/owner verifications (CIDB, SSM, insurance) are enforced on registration. All references (project ID, contract ID, etc.) are consistent across models. Milestone sums equal bid total.
- **Security:** All sensitive actions (login, payment) use HTTPS. Payment info is handled by PCI-compliant gateway. Access control: only the owner sees their projects, only bidders see bid details after login.
- **Regulatory Compliance:** Mandatory contractor info is collected. Contracts generated align with STCC guidelines ⁵. "Pay-when-paid" or similar clauses are disallowed per CIPAA ².
- **Performance:** The site loads quickly (optimised images, scripts). Payment transactions respond within a few seconds.
- **Reliability:** The escrow system accurately holds and releases funds. Failure modes (payment gateway timeout, partial upload) are handled gracefully.
- **Testing:** Each feature has unit and integration tests. Payment flows are tested with sandbox API and edge cases (insufficient funds, partial approvals). User acceptance testing ensures real users find and use core functions.
- **Monitoring and Support:** Errors are logged, admins get alerts on disputes or payment issues. Help center/FAQ covers common questions (e.g. "How does milestone payment work?").

Summary

Tukul.com aims to **revolutionize Malaysian renovations** by embedding trust and structure into the contractor marketplace. By enforcing verified credentials ⁸, milestone-based contracts, and secure escrow, Tukul ensures homeowners pay fairly for quality work, and contractors get paid promptly for

completed work. This specification builds on proven models (PilihKontraktor's trust ³, RenSecure's secure milestones ¹³) while complying with local regulations (CIDB licensing and CIPAA) to deliver a robust, transparent platform for all parties.

Sources: Malaysian CIDB (contractor registration and STCC guidance) ⁷ ⁵; RenSecure website (milestone payments, escrow model) ¹ ¹³; PilihKontraktor About page (verified contractors, trust) ³; DLA Piper (CIPAA 2012 context) ⁹; Freshfields blog (CIPAA Section 35 on pay-when-paid) ².

¹ ⁴ ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁸ RenSecure | Trusted Home Renovation Platform
<https://www.rensecure.com.my/>

² Shaping Asia's Infrastructure: "Pay when paid" clauses across Asia | Freshfields
<https://www.freshfields.com/en/our-thinking/blogs/risk-and-compliance/shaping-asias-infrastructure-pay-when-paid-clauses-across-asia-102m27s>

³ ⁶ ⁸ ¹⁷ PilihKontraktor - Ada Apps Ada Kontraktor
<https://pilihkontraktor.com/about>

⁵ ⁷ ¹² Pendaftaran Kontraktor | CIDB HQ
https://www.cidb.gov.my/eng/contractor-registration/?utm_source=chatgpt.com

⁹ ¹⁰ ¹¹ Malaysian Construction Industry Payment and Adjudication Act 2012 | DLA Piper
<https://www.dlapiper.com/en-us/insights/publications/2025/03/malaysian-construction-industry-payment-and-adjudication-act-2012>

¹⁹ What is the typical first payment to a contractor for a renovation ...
<https://www.facebook.com/groups/tukangapahariini/posts/1427731858367416/>